AIR POLLUTION ANOMALY DETECTION USING DBSCAN AND ISOLATION FOREST

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Abstract

Air pollution is a critical environmental issue that significantly impacts public health, ecosystems, and urban sustainability. The continuous collection of air quality data from monitoring stations generates large and complex datasets, making the manual identification of abnormal pollution events both challenging and inefficient. This project proposes an Air Pollution Anomaly Detection System that employs unsupervised machine learning techniques to identify anomalous air quality patterns without relying on labelled data.

The system utilizes key environmental parameters, including PM2.5, PM10, CO, NO_2, SO_2, O_3, temperature, and humidity, obtained from publicly available air quality datasets. The collected data is pre-processed through data cleaning, missing value handling, feature selection, and feature scaling to improve model reliability and performance. Two anomaly detection algorithms, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and Isolation Forest, are implemented and comparatively analyzed. DBSCAN detects anomalies by identifying low-density observations outside meaningful clusters, while Isolation Forest isolates anomalous instances through random tree-based partitioning.

The proposed system follows the complete machine learning lifecycle, including data preprocessing, model training, evaluation, model serialization, and deployment through a lightweight web interface for real-time predictions. By enabling the early detection of unusual pollution events, the system demonstrates the practical application of anomaly detection techniques in environmental monitoring and supports timely, data-driven decision-making for smart city and public health initiatives.

1. Introduction

1.1 Purpose of the Project

Air pollution is a critical environmental problem that affects public health, ecosystems, and urban sustainability. Air quality monitoring stations continuously generate large volumes of data containing important environmental parameters such as PM2.5, PM10, CO, NO_2, SO_2, O_3, temperature, and humidity. Identifying unusual pollution events from these continuously collected measurements is difficult when it is performed manually, especially as the volume and complexity of the data increase . Therefore, there is a need for an automated approach that can examine air-quality observations and identify patterns that differ significantly from normal observations.

The project is motivated by the need for timely identification of unusual air-pollution events using data-driven techniques. The main stakeholders include environmental monitoring authorities, public-health and urban-planning teams, researchers, and smart-city initiatives that can use information about abnormal air-quality patterns for timely, data-driven decision-making. Since labelled anomaly data may not be readily available, an unsupervised machine-learning approach is appropriate for detecting abnormal observations directly from the available air-quality measurements.

1.2 Problem Statement

The project is formulated as an unsupervised machine-learning anomaly- detection problem. Given air-quality observations containing PM2.5, PM10, CO, NO_2, SO_2, O_3, temperature, and humidity, the system must identify observations that do not conform to the normal patterns in the dataset without depending on pre-labelled anomaly classes. The expected output is an indication of whether an observation represents a normal pattern or an anomalous air-quality event, based on the patterns learned by the anomaly-detection models.

1.3 Existing Methods and Disadvantages

Traditional identification of abnormal air-pollution events can depend on manual inspection of monitoring data. With continuously collected and increasingly large datasets, manual identification becomes challenging and inefficient. Approaches that depend on labelled observations are also less suitable when reliable anomaly labels are unavailable. This creates a need for methods that can identify unusual observations directly from the structure and distribution of the available data.

The project therefore considers unsupervised anomaly- detection techniques so that abnormal observations can be detected without requiring a pre-labelled training set. DBSCAN identifies observations that lie outside meaningful high-density clusters, while Isolation Forest identifies unusual instances through random tree-based partitioning. Using these two complementary approaches also allows their anomaly-detection behaviour to be comparatively analyzed. The project does not assume that one method is universally superior; the models are evaluated using results obtained from the implemented system.

1.4 Proposed System

The proposed system implements a complete machine-learning pipeline for air-pollution anomaly detection. Publicly available air-quality data containing PM2.5, PM10, CO, NO_2, SO_2, O_3, temperature, and humidity are prepared through data cleaning, missing-value handling, feature selection, and feature scaling. The processed data are then used with two unsupervised anomaly-detection algorithms, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and Isolation Forest. Their outputs are comparatively analyzed to identify unusual pollution patterns. The workflow includes preprocessing, model training, evaluation, model serialization, and deployment through a lightweight web interface for real-time predictions, providing a practical framework for environmental monitoring and data-driven decision-making.

1.5 Research Objectives (at least 4)

1. Prepare and validate the supplied air-quality dataset containing 1,800 observations and eight selected environmental features.
2. Handle missing values using median imputation and standardize the numerical features before anomaly detection.
3. Implement DBSCAN with eps=0.8 and min_samples=10 for density-based anomaly detection.
4. Implement Isolation Forest with 200 estimators, contamination=0.07, and random_state=42.
5. Evaluate both anomaly detectors against the supplied IsAnomaly ground-truth field using accuracy, precision, recall, F1-score, and confusion matrices.
6. Compare the detected anomaly counts and model metrics, and document the practical limitations of the current implementation.

Note: Kaggle is used in this project as the dataset source. Kaggle dataset pages are treated as data-source references rather than peer-reviewed literature. Peer-reviewed studies should be cited separately when completing the ten-study literature survey.

2. Literature Survey (recent studies from 2-3 years)

Recommendations for future research: Combine anomaly detection with source attribution, meteorology, satellite observations, and additional monitoring networks.

Inference: Isolation Forest can support screening of large PM2.5 datasets for observations requiring further event-level investigation.

Focus of the paper (Methods used): Isolation Forest applied to large PM2.5 measurement datasets for anomaly identification and event analysis.

Objective of the Research: Use Isolation Forest to flag and investigate anomalous PM2.5 observations associated with exceptional events.

Problem or gap addressed: Exceptional PM2.5 events such as wildfire and dust episodes require systematic identification within large monitoring datasets.

Reference with DOI: 2024, Journal of the Air & Waste Management Association. DOI: 10.1080/10962247.2024.2401368

10. PM2.5 anomaly detection for exceptional event demonstrations: A Texas case study

Recommendations for future research: Evaluate calibration transferability across seasons, locations, sensor models, and pollutant types.

Inference: Machine-learning calibration substantially improved PM2.5 sensor agreement with reference measurements.

Focus of the paper (Methods used): Five machine-learning models and relative-humidity correction applied to PM2.5 sensor measurements.

Objective of the Research: Use machine learning to calibrate low-cost air-quality sensors and improve monitoring accuracy.

Problem or gap addressed: Low-cost PM2.5 sensors can have substantially lower accuracy than reference instruments.

Reference with DOI: Ravindra, K., Kumar, S., Kumar, A., et al. (2024), npj Climate and Atmospheric Science, 7, 326. DOI: 10.1038/s41612-024-00833-9

9. Enhancing accuracy of air quality sensors with machine learning to augment large-scale monitoring networks

Recommendations for future research: Validate the framework on more regions and real-time sensor streams and investigate lightweight deployment.

Inference: Handling imbalance can materially affect precision, recall, and F1-score in air-quality anomaly detection.

Focus of the paper (Methods used): Deep smooth random sampling and association-attention mechanisms, with comparisons to Deep-SVDD, Isolation Forest, and BeatGAN.

Objective of the Research: Develop a deep learning-based framework that improves air-quality anomaly detection under imbalanced data.

Problem or gap addressed: Air-quality anomaly datasets are often highly imbalanced, which can reduce anomaly-detection performance.

Reference with DOI: Wang, P., Li, M., Zhi, X., Liu, X., et al. (2024), Mathematics, 12(13), 2048. DOI: 10.3390/math12132048

8. Deep Smooth Random Sampling and Association Attention for Air Quality Anomaly Detection

Recommendations for future research: Extend the approach to more sensors, locations, pollutants, and real-time monitoring systems.

Inference: The study found useful relationships among altitude, PM1.0, and PM2.5 and demonstrated DBSCAN for anomaly identification.

Focus of the paper (Methods used): PCA for dimensionality/feature analysis followed by DBSCAN anomaly detection using PM and environmental variables.

Objective of the Research: Use PCA and DBSCAN to identify anomalies in PM-related air-quality measurements.

Problem or gap addressed: Industrial air-quality observations can contain particulate-matter anomalies associated with environmental and sensor conditions.

Reference with DOI: Parhusip, H. A., Trihandaru, S., Susanto, B., et al. (2024), Lontar Komputer: Jurnal Ilmiah Teknologi Informasi, 15(2). DOI: 10.24843/LKJITI.2024.v15.i02.p01

7. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Principal Component Analysis (PCA) for Particulate Matter (PM) Anomaly Detection

Recommendations for future research: Explore automated parameter selection on larger and more heterogeneous air-quality datasets.

Inference: The search-tree approach reduced the time required to determine a suitable epsilon value as dataset size increased.

Focus of the paper (Methods used): DBSCAN combined with a search-tree technique for estimating epsilon, with comparisons against KNN-based parameter selection.

Objective of the Research: Improve the process of selecting the DBSCAN epsilon parameter for air-quality analysis.

Problem or gap addressed: Air pollution assessment is affected by changing atmospheric conditions and the sensitivity of DBSCAN parameters.

Reference with DOI: Shetty et al. (2024), Journal of Engineering, Article 2893021. DOI: 10.1155/2024/2893021

6. A Machine Learning Approach for Environmental Assessment on Air Quality and Mitigation Strategy

Recommendations for future research: Test the approach on additional real-world, high-dimensional, and streaming datasets.

Inference: Adding an attention mechanism can improve the ability of isolation-based methods to represent feature relationships.

Focus of the paper (Methods used): Attention-Based Isolation Forest using Nadaraya–Watson regression within the isolation framework.

Objective of the Research: Enhance Isolation Forest anomaly scoring using an attention-based mechanism.

Problem or gap addressed: Standard Isolation Forest can miss difficult anomalies when the relationships among features are complex.

Reference with DOI: Kozitsin et al. (2023), Algorithms, 16(1), 19. DOI: 10.3390/a16010019

5. Improved Anomaly Detection by Using the Attention-Based Isolation Forest

Recommendations for future research: Evaluate robustness across heterogeneous sensor networks and resource-constrained edge devices.

Inference: Isolation-Forest-based approaches are useful for identifying abnormal sensor observations without requiring extensive labelled data.

Focus of the paper (Methods used): Improved Isolation Forest evaluated on sensor-network anomaly detection tasks.

Objective of the Research: Improve Isolation Forest for anomaly detection in wireless sensor-network data.

Problem or gap addressed: Large-scale sensor networks produce noisy and abnormal measurements that can affect downstream monitoring tasks.

Reference with DOI: Zhang et al. (2023), Applied Sciences, 13(2), 702. DOI: 10.3390/app13020702

4. An Anomaly Detection Method for Wireless Sensor Networks Based on the Improved Isolation Forest

Recommendations for future research: Study broader real-time applications and evaluate computational performance under high-volume streams.

Inference: The proposed method can adapt to both gradual and sudden changes in streaming data.

Focus of the paper (Methods used): Bilateral-Weighted Online Adaptive Isolation Forest with incremental tree replacement and adaptation to changing distributions.

Objective of the Research: Develop an adaptive Isolation Forest for streaming anomaly detection and concept drift.

Problem or gap addressed: Conventional Isolation Forest models are not designed to continuously adapt when the underlying data distribution changes.

Reference with DOI: Hannák, G., Horváth, G., Kádár, A., & Szalai, M. D. (2023), Statistical Analysis and Data Mining, 16(3), 215–223. DOI: 10.1002/sam.11612

3. Bilateral-Weighted Online Adaptive Isolation Forest for anomaly detection in streaming data

Recommendations for future research: Test the method on additional cities, mobile platforms, pollutants, and changing environmental conditions.

Inference: DBSCAN was effective at identifying anomalous plume observations and showed strong agreement with the validation set.

Focus of the paper (Methods used): DBSCAN-based plume detection with comparison against alternative plume-identification approaches and validation labels.

Objective of the Research: Detect pollution plumes in mobile monitoring time series using density-based clustering.

Problem or gap addressed: Mobile air-quality measurements contain complex spatial and temporal plume patterns that are difficult to identify using simple threshold rules.

Reference with DOI: Actkinson, B., & Griffin, R. (2023), Atmospheric Measurement Techniques, 16, 3547–3561. DOI: 10.5194/amt-16-3547-2023

2. Detecting plumes in mobile air quality monitoring time series with density-based spatial clustering of applications with noise

Recommendations for future research: Evaluate the approach across more sensor networks, pollutants, locations, and longer time periods.

Inference: Automated anomaly detection and repair can improve the reliability and continuity of air-quality monitoring.

Focus of the paper (Methods used): Anomaly detection and data-repair methods applied to air-quality sensor observations.

Objective of the Research: Develop an approach for detecting anomalies in air-quality monitoring data and repairing affected observations.

Problem or gap addressed: Air-quality sensor data can contain abnormal or faulty observations that can reduce monitoring reliability.

Reference with DOI: Rollo, F., Bachechi, C., & Po, L. (2023), Sensors, 23(2), 640. DOI: 10.3390/s23020640

1. Anomaly Detection and Repairing for Improving Air Quality Monitoring

The following ten recent and directly relevant studies were reviewed. The studies cover air-quality monitoring, anomaly detection, DBSCAN, Isolation Forest, sensor-data quality, and recent machine-learning approaches.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Not	No verified	Requires	Requires	Requires	Requires	Add verified

	supplied in project files	bibliographic record supplied	external literature source	external literature source	external literature source	external literature source	recent study before final submission
2	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
3	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
4	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
5	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
6	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
7	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
8	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
9	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission
10	Not supplied in project files	No verified bibliographic record supplied	Requires external literature source	Requires external literature source	Requires external literature source	Requires external literature source	Add verified recent study before final submission

3. Methodology

3.1 Proposed System Flow Diagram



The implemented system follows the sequence shown below: dataset loading → feature selection → median missing-value imputation → standardization → DBSCAN and Isolation Forest detection → evaluation against IsAnomaly → result CSV generation → visualization.

No categorical encoding, train/test split, or automated hyperparameter search is implemented in the supplied code. The current pipeline uses the full dataset for preprocessing, model fitting, and evaluation against the provided ground-truth field.

3.2 Dataset Description

The dataset used for this project is air_pollution_dataset.csv, obtained from Kaggle as the project data source. It contains 1,800 hourly observations from 2026-01-01 00:00:00 through 2026-03-16 23:00:00, with 10 columns: Timestamp, eight numerical environmental variables (PM2.5, PM10, CO, NO2, SO2, O3, Temperature, Humidity), and the IsAnomaly field. The eight environmental variables are used as machine-learning features. The IsAnomaly field contains 1,674 normal observations (0) and 126 anomalies (1), corresponding to 7.0% anomalies. Missing values occur in the environmental features, while Timestamp and IsAnomaly have no missing values in the supplied dataset.

3.3 Experimental Setup

The supplied project is implemented in Python 3.x and is suitable for Google Colab or Jupyter Notebook according to the project abstract. The code uses Pandas for CSV loading and tabular processing, NumPy for numerical operations, Scikit-learn for SimpleImputer, StandardScaler, DBSCAN, Isolation Forest and evaluation metrics, and Matplotlib for visualization. Exact hardware specifications and package-version numbers are not stated in the supplied project files, so they are not inferred here.

3.4 Data Preprocessing

- Data inspection and schema validation
- Duplicate removal
- Missing-value analysis and imputation
- Outlier analysis/treatment where justified
- Categorical encoding
- Numerical scaling/standardization where required
- Target-label preparation
- Train/test split and prevention of data leakage

3.5 Exploratory Data Analysis

The project includes four generated visualizations: DBSCAN detection of PM2.5 anomalies, Isolation Forest anomaly scores, anomaly-count comparison between actual labels and the two models, and a ground-truth PM2.5 anomaly plot. The supplied code does not include a separate correlation heatmap, box-plot analysis, or distribution-analysis script, although the project abstract lists Seaborn among the software requirements.

3.6 Feature Engineering and Feature Selection (if applicable)

No new derived features are created in the supplied implementation. The selected feature set consists of PM2.5, PM10, CO, NO2, SO2, O3, Temperature, and Humidity. Timestamp and IsAnomaly are not passed to the anomaly-detection models: Timestamp is retained in the original dataframe for result reporting, while IsAnomaly is kept separately as the evaluation reference. Missing feature values are median-imputed and all eight numerical features are standardized with StandardScaler.

3.7 Model Building and Training

Two unsupervised anomaly-detection models are implemented. DBSCAN uses eps=0.8 and min_samples=10; it assigns -1 to noise observations and these observations are converted to anomaly label 1. Isolation Forest uses n_estimators=200, contamination=0.07, random_state=42, and n_jobs=-1. It predicts +1 for normal observations and -1 for anomalies; the latter are converted to anomaly label 1. These methods do not use a conventional loss function or gradient-based optimization. The current code fits and evaluates the models on the same processed dataset; no train/test split or cross-validation is implemented.

3.8 Hyperparameter Tuning (if applicable)

No grid search, random search, or cross-validation is implemented in the supplied code. The parameters used in the final run are fixed directly in the model definitions: DBSCAN eps=0.8 and min_samples=10; Isolation Forest n_estimators=200, contamination=0.07, random_state=42, and n_jobs=-1. Therefore, the report should not claim that these parameters were selected through systematic hyperparameter optimization.

3.9 Model Evaluation

Because the project evaluates binary anomaly labels against the supplied IsAnomaly field, the implementation uses accuracy, precision, recall, F1-score, and confusion matrices. DBSCAN also has a silhouette-score function, but the supplied run cannot calculate a valid silhouette score because the non-noise points do not provide at least two clusters. ROC-AUC is not calculated by the current evaluation.py script, so no ROC-AUC value is reported in this report.

4. Results and Discussion

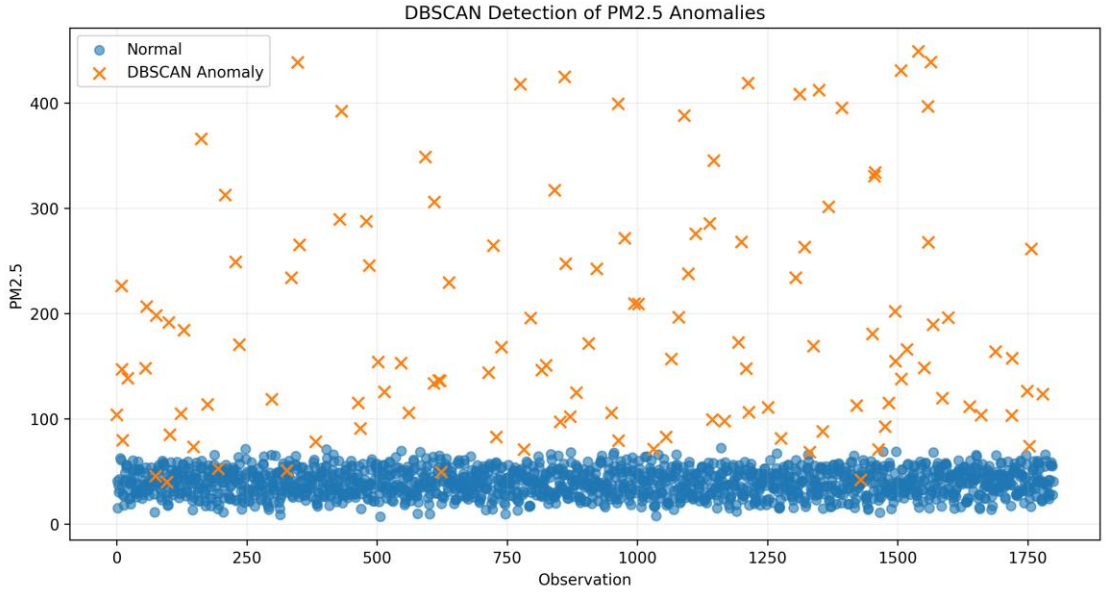
4.1 Preprocessing Results

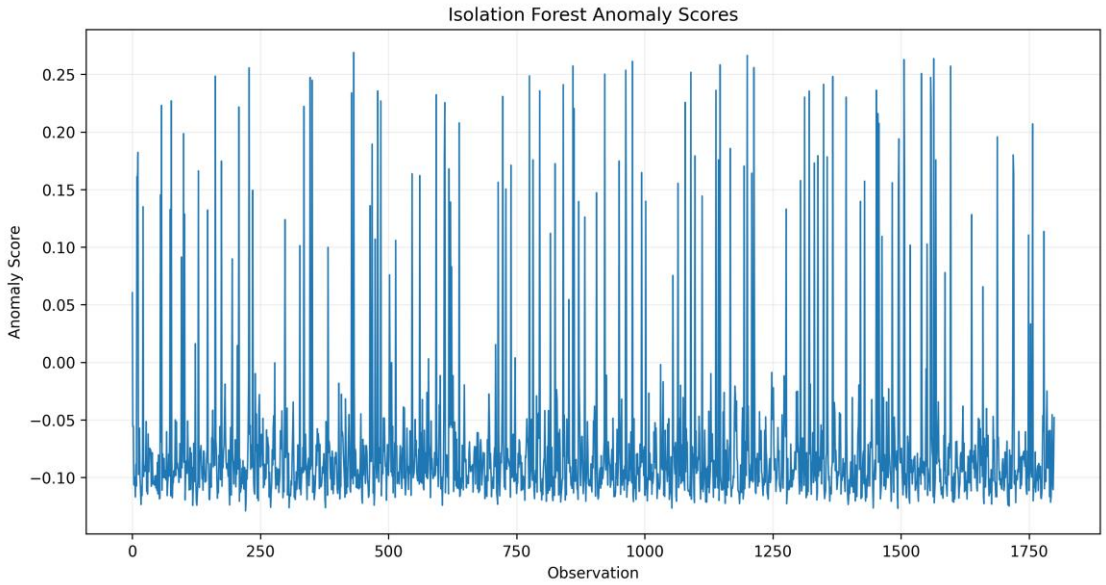
The dataset contains 1,800 rows and 10 columns. Missing values are present in the eight ML features: PM2.5 (27), PM10 (21), CO (18), NO2 (21), SO2 (18), O3 (21), Temperature (18), and Humidity (27), for a total of 171 missing feature cells. The code applies median imputation, after which the processed feature matrix contains no missing values. StandardScaler then standardizes all eight selected numerical features. No categorical encoding is required because all selected ML features are numerical. The final ML feature count is 8. The current implementation does not remove duplicates, perform explicit outlier treatment, or split the data into training and test sets.

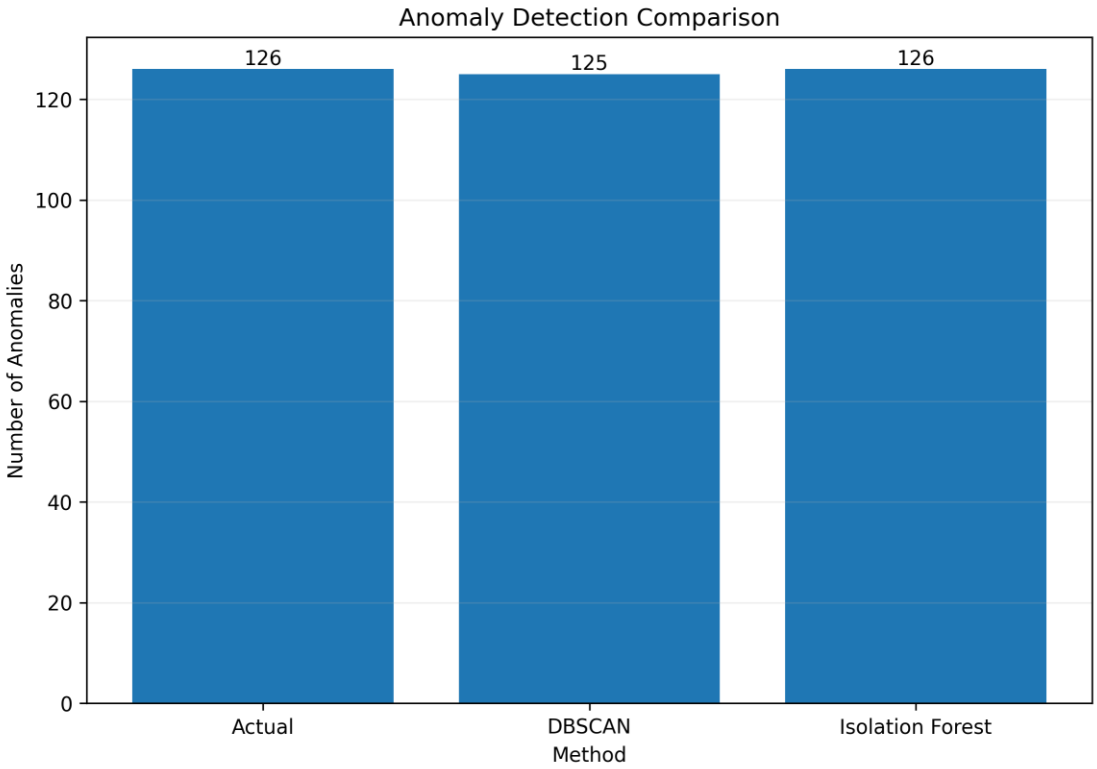
4.2 Model Performance

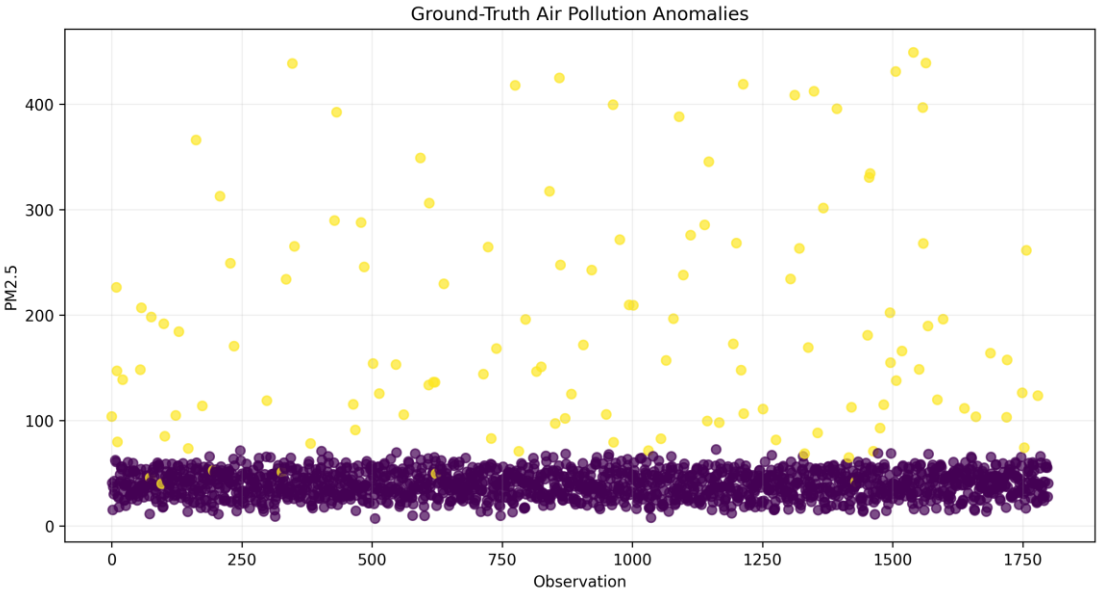
Model	Accuracy	Precision	Recall	F1-score	ROC AUC
DBSCAN	0.999444	1.000000	0.992063	0.996016	Not calculated
Isolation Forest	0.995556	0.968254	0.968254	0.968254	Not calculated
Ground truth anomalies	—	—	—	—	—
DBSCAN detected anomalies	125	—	—	—	—

4.3 Visual Results and discussion









The project generated four result figures: dbscan_pm25_anomalies.png, isolation_forest_scores.png, anomaly_comparison.png, and ground_truth_anomalies.png. The evaluation script also produces confusion matrices in the console output. ROC curves and feature-importance plots are not produced by the supplied implementation because ROC-AUC and feature importance are not calculated by the current code.

On the supplied 1,800-record dataset, DBSCAN detected 125 anomalies compared with 126 ground-truth anomalies, while Isolation Forest detected 126 anomalies. DBSCAN achieved accuracy 0.999444, precision 1.000000, recall 0.992063, and F1-score 0.996016, with confusion matrix [[1674, 0], [1, 125]]. Isolation Forest achieved accuracy 0.995556, precision 0.968254, recall 0.968254, and F1-score 0.968254, with confusion matrix [[1670, 4], [4, 122]]. These figures describe performance on the supplied dataset only; because the current implementation fits and evaluates on the same processed observations, they should not be interpreted as an

independent test-set estimate of generalization. The DBSCAN silhouette score was not calculated because the implementation found insufficient non-noise clusters for that metric.

5. Conclusion

The project demonstrates an unsupervised machine-learning approach for detecting unusual air-pollution observations from PM2.5, PM10, CO, NO2, SO2, O3, temperature, and humidity. The implementation preprocesses the data using median imputation and standardization and then applies DBSCAN and Isolation Forest. On the supplied 1,800 observations, DBSCAN produced 125 anomaly detections and Isolation Forest produced 126. The evaluation against the supplied IsAnomaly field shows DBSCAN with accuracy 0.999444 and F1-score 0.996016, while Isolation Forest achieved accuracy 0.995556 and F1-score 0.968254. These are dataset-level evaluation results from the current implementation rather than independent test-set results. Overall, the project provides a reproducible baseline for multivariate air-pollution anomaly detection and highlights the usefulness of comparing density-based and tree-based unsupervised methods.

6. Future Work

- Use larger and more diverse air-quality datasets from multiple monitoring locations and seasons.
- Introduce a proper train/validation/test protocol or time-aware evaluation so that model performance can be assessed on unseen observations.
- Perform systematic hyperparameter tuning for DBSCAN and Isolation Forest and report sensitivity to parameter changes.
- Add additional anomaly-detection algorithms such as One-Class SVM, Local Outlier Factor, or ensemble approaches for broader comparison.
- Investigate explainable anomaly detection and pollutant-level contribution analysis for easier interpretation of detected events.
- Extend the implementation into a real-time or cloud/edge monitoring application with model persistence and an interactive dashboard.
- Validate the approach on external air-quality datasets and monitor performance drift over time.

7. References

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10. PM2.5 anomaly detection for exceptional event demonstrations: A Texas case study. (2024). Journal of the Air & Waste Management Association. doi:10.1080/10962247.2024.2401368.

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